1 Trade policy (8 points)

Tariffs frequently impact the flow of goods internationally. In this section, you will draw on data from one of these tariffs, investigating how these disruptions impact exports. For the next five questions, you will use the data posted on Canvas as *trade_policy.csv*. This data contains several variables: *post* is a binary variable indicating the tariff has been introduced (post = 1, pre = 0); *treat* is a binary variable that equals 1 if a country is "impacted" by the tariff barrier (i.e., a regulation increasing the cost of exports) and 0 if a country is "not impacted"; *air_conditioners_pc* is the number of air conditioners exported per capita by a given country in a year.

1. Use a difference-in-differences design to estimate the impact of the tariff on exports of air conditioners. Regress *air_conditioners_pc* on *post*, *treat*, and the interaction of *post* and *treat*. Which coefficient reveals the change in exports after the tariff was introduced? Explain and interpret the result. (4 points)

Assuming the regression equation is:

$$air_conditioners_pc = \beta_0 + \beta_1(treat) + \beta_2(post) + \beta_3(post \times treat) + \epsilon$$

Explanation: The coefficientβ_3 is the DiD estimate. It compares how much exports changed for the impacted countries after the tariff, relative to the baseline change seen in the unimpacted countries during that same time window.

Interpretation: This coefficient represents the causal effect of the tariff on air conditioner exports. Specifically, it tells us how many more (or fewer) air conditioners per capita the impacted countries exported purely due to the tariff, compared to what they would have exported had the tariff not been introduced.

2. A friend is suggesting that you should considering taking the log of your outcome, *air_conditioners_pc*. What are some potential concerns regarding this suggestion? Describe two. (2 points)

The Issue of Zero-Valued Observations: In instances where a country records zero air conditioner exports, a logarithmic transformation fails because ln (0) is undefined. Consequently, researchers must either omit these data points—which introduces sample selection bias—or apply ad-hoc transformations such as ln(Y+1) which fundamentally alter the economic interpretation of the estimates.

Violating Parallel Trends Assumption: The DiD design depends strictly on the parallel trend's assumption. If the treatment and control groups have parallel trends in *levels* (absolute numbers), they will mathematically not have parallel trends in *logs* (proportional/percentage changes), unless their baseline levels are identical. Transforming logs could artificially create a violation of the foundational DiD assumption.

3. The difference-in-differences design is valid when treatment and control have common trends. What is a common trend? Write down the definition. Next, give an example (about this case) when this assumption likely violated. One or two sentences would suffice. (2 points)

Definition: The common trends (or parallel trends) assumption states that in the absence of the treatment (the tariff), the average outcomes for the treatment group and the control group would have followed parallel paths over time.

Violation Example: This assumption would likely be violated if, right before the tariff was introduced, the impacted countries started providing heavy government subsidies to their domestic appliance manufacturing sector, while the "not impacted" countries did not. This means the impacted countries' exports would have surged anyway, making the control group an invalid baseline for comparison.

1. Consider the variable *ever_medicaid*, which reports whether someone actually enrolled in Medicaid coverage since the lottery. Did everyone who won the lottery take up Medicaid? Calculate the mean of *ever_medicaid* among those who won the lottery.
   Are there some people who lost the lottery who take up Medicaid? Calculate the mean of *ever_medicaid* for those that lost the lottery.

   Now, recall the key regression from Problem Set #1:

$$Y = \beta_0 + \beta_1 treated + \epsilon$$

   We have been interpreting the coefficient on *treated* as the causal effect of winning the Medicaid lottery. Is this likely to be the same as the causal effect of actually receiving Medicaid? (2 points)

Mean of ever_medicaid among lottery winners (treated=1): 0.4256 (≈42.56%).

⇒ Not everyone who won the lottery enrolled in Medicaid.

Mean of ever_medicaid among lottery losers (treated=0): 0.1857 (≈18.57%).

⇒ Yes, some people who lost the lottery still ended up on Medicaid.

No, not in general. Because take-up is imperfect (noncompliance), the regression with treated identifies an ITT (intention-to-treat) effect: the causal effect of being offered the chance to enroll, not the effect of enrollment itself.

A tibble: 2 × 3

treated <dbl>	n <int>	mean_ever_medicaid <dbl>
0	5842	0.1857241
1	6387	0.4255519

2. Estimate the following "First-stage" regression:

$$ever_medicaid = \alpha_0 + \alpha_1 treated + \alpha_2 numhh_list + e \tag{1}$$

Interpret the regression coefficient on α_1. (2 points)

Estimated $\alpha \approx 0.2443$

Interpretation: Winning the lottery increases the probability of ever enrolling in Medicaid by about 24.4 percentage points, holding numhh_list constant.

3. Is *treated* likely to be a valid instrument for *ever_medicaid* in terms of the three key assumptions: instrument relevance, independence assumption and the exclusion restriction? If there is a specific statistical test that helps validate the plausibility of each assumption, please refer to this evidence (3 points)

Very likely satisfied. In the first-stage regression, α_1 is large and highly significant; the first-stage F-statistic for treated is about 915 （$\gg 10$）, strongly rejecting weak instruments.

Plausible because the lottery assignment is designed to be random. In practice, you support this by balance tests: check whether baseline covariates (e.g., sex/age/race/education/health baseline) differ by treated. Because winning probabilities can vary with numhh_list, it's common to check balance conditional on numhh_list (as my code does).

Exclusion restriction: Hard to test directly (especially with a single instrument). It must be argued from design: winning should affect outcomes only through Medicaid take-up. With one instrument there is no overidentification test; you can only use indirect evidence like placebo/falsification tests (e.g., no effect on pre-lottery outcomes, if available).

```
---- First-stage partial F for treated (robust) ----
F = 915.413
p = 5.583018e-194

---- Check (single restriction): F = t^2 ----
t = 30.25579
F = t^2 = 915.413
```

4. One possible way to estimate the effect of receiving health insurance is the ratio of the coefficient on the instrument in the reduced form model to the coefficient on the instrument in the first-stage regression. Let's again focus on the outcome *count_visit_dr*. Recall, the reduced form is the regression we've already been running in Problem Sets #1 and #3:

$$count_visit_dr = \beta_0 + \beta_1 treated + \beta_2 numhh_list + \epsilon \tag{2}$$

Calculate and report the ratio of the reduced form to the first stage. Note that technically, we should use the same number of observations in our first-stage and reduced form, so you should calculate the first stage excluding rows with missing values of *count_visit_dr*. Interpret your result (2 points)

Using the same sample for both regressions (dropping missing count_visit_dr, N = 12,158):

Reduced form coefficient on treated:$\beta_1(hat) = 0.5935$

First-stage coefficient on treated:$\alpha_1(hat) = 0.2435$

$$\beta_1/\alpha_1 = 2.4372$$

Interpretation: Under the IV assumptions, this ratio is the Wald/IV (LATE) estimate of the effect of *enrolling in Medicaid* on count_visit_dr. It implies that Medicaid enrollment increases doctor visits by about 2.44 visits for compliers (those whose enrollment is affected by winning the lottery), over the outcome's measurement period.

```
N used in both regressions = 12158

Reduced form coef on treated (beta1) = 0.5935102
First stage  coef on treated (alpha1) = 0.243519
Ratio (beta1/alpha1) = 2.437223
```

5. Let's next estimate the IV model via two-stage least squares (2SLS). Manually estimate the two stages:

   (a) **First-stage:** Run the regression $ever_medicaid = \alpha_0 + \alpha_1 treated + \alpha_2 numhh_list + e$ and compute predicted values: $\widehat{ever_medicaid} = \widehat{\alpha}_0 + \widehat{\alpha}_1 treated + \widehat{\alpha}_2 numhh_list$

   (b) **Second-stage:** Run the regression $count_visit_dr = \beta_0 + \beta_1 \widehat{ever_medicaid} + \beta_2 numhh_list +$

Again, you should only estimate the first stage on the subset of the data where *count_visit_dr* is not missing. How does your result compare to the previous estimate? You can now also conduct inference–is the IV estimate statistically significant? (3 points)

$\alpha_1(hat) = 0.2435 SE \approx 0.0082$

Winning the lottery increases Medicaid take-up by 24.35 pp conditional on numhh_list.

$$\beta_1(hat) = 2.4372$$

The IV estimate is statistically significant (p≈0.006).

```
---- Key IV result ----
beta_IV (effect of Medicaid on count_visit_dr) = 2.437223
robust SE (HC1)                                = 0.8850557
p-value                                        = 0.005900407
```

6. Now, we will try including all the baseline characteristics as controls. However, unlike in Problem Set #3, let's account for a non-linearity in age by including a quadratic term, age^2.

   Thus, the full-set of controls become:
   $numhh_list, female, age, age^2, race_white, hs_degree, college_degree, health_baseline.$

   Re-estimate the 2SLS model including these controls and compare the size of the coefficient on $ever_medicaid$ and the standard error to the results in part (6). (2 points)

N=12125

$$\beta_{ever-medicaid}(hat) = 2.4247$$

SE = 0.8636（p≈0.0050）

After adding the full set of baseline controls (including *age2)*, the IV coefficient on ever_medicaid is essentially unchanged （2.437 → 2.425）, while the standard error becomes slightly smaller (0.881 → 0.864). This suggests the estimate is robust to adding controls, and precision improves modestly. The IV estimate remains statistically significant.

```
   Estimate  Std. Error    Pr(>|t|)
2.437223144 0.880641035 0.005656383
   Estimate  Std. Error    Pr(>|t|)
2.424652122 0.863600397 0.004999059
[1] 12158
[1] 12125
```

7. Let's calculate the IV estimates for all the endline outcomes now, including the full set of controls.

   To make things easier, you are welcome to use the IV *R* command, *ivreg*. You will first need to install the *R* package *AER*. Do this by typing into *R*: install.packages("AER"). Then load the package: library("AER").

   For each of the endline outcomes, run the IV regressions including the controls. Fill in the table below with your results. Discuss your findings, including their precision. (2 points)

	(1)	(2)
Endline outcome	IV estimate	S.E.
visit_dr		
visit_er		
out_of_pocket_spend		
health_score		
happy		

-

	IV_estimate <dbl>	SE <dbl>	N <dbl>
visit_dr	2.4247	0.8636	12125
visit_er	0.0432	0.1431	12142
out_of_pocket_spend	-283.0383	87.9031	12112
health_score	-0.0023	0.0040	9392
happy	0.0265	0.0314	12170

5 rows

Doctor visits increase significantly (IV ≈ +2.425, SE 0.864, p≈0.005).

ER visits show no statistically significant effect (estimate near zero, relatively imprecise).

Out-of-pocket spending decreases significantly (IV ≈ −283, SE 87.9, p≈0.001), fairly precise.

No significant effects on health_score or happy; estimates are small and/or imprecise (especially health_score due to smaller N).

#1

```{r }
setwd('/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw4')
library(tidyverse)
library(lmtest)
library(sandwich)

ohie <- read_csv("OHIE.csv")

means_tab <- ohie %>%
  group_by(treated) %>%
  summarise(
    n = n(),
    mean_ever_medicaid = mean(ever_medicaid, na.rm = TRUE),
    .groups = "drop"
  )
means_tab
```

#2

```{r }
setwd('/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw4')
library(tidyverse)
library(lmtest)
library(sandwich)
ohie <- read_csv("OHIE.csv")
fs <- lm(ever_medicaid ~ treated + numhh_list, data = ohie)
summary(fs)
coeftest(fs, vcov = vcovHC(fs, type = "HC1"))
waldtest(fs, . ~ numhh_list, vcov = vcovHC(fs, type = "HC1"))

```

#3

```{r }
setwd('/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw4')
library(tidyverse)
library(lmtest)
library(sandwich)

ohie <- read_csv("OHIE.csv")

fs <- lm(ever_medicaid ~ treated + numhh_list, data = ohie)
coeftest(fs, vcov = vcovHC(fs, type = "HC1"))
wald_out <- waldtest(fs, . ~ numhh_list, vcov = vcovHC(fs, type = "HC1"))
print(wald_out)
F_value <- wald_out$F[2]
p_value <- wald_out$`Pr(>F)`[2]

cat("\n---- First-stage partial F for treated (robust) ----\n")
cat("F =", F_value, "\n")
cat("p =", p_value, "\n")

ct <- coeftest(fs, vcov = vcovHC(fs, type = "HC1"))
t_treated <- ct["treated","t value"]
F_from_t2 <- t_treated^2

cat("\n---- Check (single restriction): F = t^2 ----\n")
cat("t =", t_treated, "\n")
cat("F = t^2 =", F_from_t2, "\n")
```

#4

```{r }
setwd('/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw4')
library(tidyverse)
library(lmtest)
library(sandwich)
ohie <- read_csv("OHIE.csv")
d4 <- ohie %>% filter(!is.na(count_visit_dr))
rf <- lm(count_visit_dr ~ treated + numhh_list, data = d4)
fs <- lm(ever_medicaid ~ treated + numhh_list, data = d4)
beta1  <- coef(rf)["treated"]
alpha1 <- coef(fs)["treated"]
ratio <- beta1 / alpha1
cat("N used in both regressions =", nrow(d4), "\n\n")
cat("Reduced form coef on treated (beta1) =", beta1, "\n")
cat("First stage  coef on treated (alpha1) =", alpha1, "\n")
cat("Ratio (beta1/alpha1) =", ratio, "\n")

```

#5

```{r }
library(tidyverse)
library(lmtest)
library(sandwich)
ohie <- read.csv("OHIE.csv")
d <- subset(ohie, !is.na(count_visit_dr))
fs <- lm(ever_medicaid ~ treated + numhh_list, data = d)
summary(fs)
d$ever_hat <- predict(fs, newdata = d)
ss_manual <- lm(count_visit_dr ~ ever_hat + numhh_list, data = d)
summary(ss_manual)
coef(ss_manual)["ever_hat"]
# 和第4题比率核对
rf <- lm(count_visit_dr ~ treated + numhh_list, data = d)
ratio <- coef(rf)["treated"] / coef(fs)["treated"]
ratio
nrow(d)  # 样本量核对
if (!require(AER)) install.packages("AER")
library(AER)

iv <- ivreg(count_visit_dr ~ ever_medicaid + numhh_list | treated + numhh_list, data = d)
summary(iv)
if (!require(sandwich)) install.packages("sandwich")
if (!require(lmtest)) install.packages("lmtest")
library(sandwich); library(lmtest)

coeftest(iv, vcov. = vcovHC(iv, type = "HC1"))
```

#6

```{r }
setwd('/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw4')
ohie <- read.csv("OHIE.csv")
d <- subset(ohie, !is.na(count_visit_dr))
d$age2 <- d$age^2
if (!require(AER)) install.packages("AER")
library(AER)
iv_base <- ivreg(
  count_visit_dr ~ ever_medicaid + numhh_list |
    treated + numhh_list,
  data = d
)
summary(iv_base)
iv_full <- ivreg(
  count_visit_dr ~ ever_medicaid + numhh_list + female + age + age2 +
    race_white + hs_degree + college_degree + health_baseline |
    treated + numhh_list + female + age + age2 +
    race_white + hs_degree + college_degree + health_baseline,
  data = d
)
summary(iv_full)
b_base <- coef(summary(iv_base))["ever_medicaid", c("Estimate","Std. Error","Pr(>|t|)")]
b_full <- coef(summary(iv_full))["ever_medicaid", c("Estimate","Std. Error","Pr(>|t|)")]
b_base
b_full
nobs(iv_base)
nobs(iv_full)
```

```r
setwd('/Users/huangyichen/Desktop/UCHI-QUARTER/2026winter/STATS 2/homework/hw4')
ohie <- read.csv("OHIE.csv")
if (!require(AER)) install.packages("AER")
library(AER)
ohie <- read.csv("OHIE.csv")
ohie$age2 <- ohie$age^2
controls <- c("numhh_list","female","age","age2",
              "race_white","hs_degree","college_degree","health_baseline")
outcomes <- c(
  visit_dr = "count_visit_dr",
  visit_er = "count_visit_er",
  out_of_pocket_spend = "out_of_pocket_spend",
  health_score = "health_score",
  happy = "happy"
)
run_iv <- function(y) {
  f <- as.formula(
    paste0(y, " ~ ever_medicaid + ", paste(controls, collapse = " + "),
           " | treated + ", paste(controls, collapse = " + "))
  )
  m <- ivreg(f, data = ohie)
  s <- summary(m)
  c(
    IV_estimate = coef(s)["ever_medicaid","Estimate"],
    SE = coef(s)["ever_medicaid","Std. Error"],
    N = nobs(m)
  )
}
res <- t(sapply(outcomes, run_iv))
res <- as.data.frame(res)
res$IV_estimate <- round(res$IV_estimate, 4)
res$SE <- round(res$SE, 4)
```